

MATH 3042 — Midterm Exam- R Question

Exam instructions:

Time: 90 minutes

Please include screenshots of your RStudio **code**, the corresponding **output**, and any necessary **explanations** in this document. When you have completed the exam, save the document as a **PDF** and submit it to the Learning Hub (Lab) using the Assignment tab.

You are permitted to use AI tools and previously prepared R code. However, this exam assesses your statistical understanding. Regardless of the resources you use, you are responsible for verifying your data, code, and results to ensure they are correct.

Please do not communicate with other students in any form during the exam.

Please do not share this exam with other students.

Good luck!

Data Collection Overview

The Google Play Store dataset is a publicly available snapshot of app listings scraped from Google's Play Store. It is available on Kaggle and is widely used for data analysis and machine learning practice. Key characteristics of this dataset include:

- Self-selected listings — apps are published voluntarily by developers; the dataset does not represent all possible apps equally.
- English-language focus — the majority of app names, descriptions, and categories are in English, over-representing apps targeting English-speaking markets.
- Point-in-time snapshot — the data was collected at a specific point in time (mid-2018); app ratings, reviews, and install counts have changed since.
- Developer-and-consumer facing — the dataset includes both productivity tools and entertainment apps, spanning a very wide range of use cases.
- Contains approximately 10,000 app listings across 13 columns.
- Biased toward apps with reviews — apps with no reviews or very low installs may be under-represented or contain missing values.

For this question, you will use the full dataset as downloaded from Kaggle¹. The file is named GooglePlayStore.xlsx.

Column Descriptions

¹ L. Gupta, "Google Play Store Apps," Feb 2019. [Online]. Available: <https://www.kaggle.com/lava18/google-play-store-apps>

Column	Type	Description
App	Character	Name of the application.
Category	Character	The category the app belongs to (e.g., GAME, TOOLS, FAMILY). There are 34 unique categories.
Rating	Character*	Average user rating out of 5.0. Contains missing values (NAs) for apps with no ratings.
Reviews	Character*	Number of user reviews. Stored as character — must be converted to numeric.
Size	Character	App size on disk (e.g., '19M', '2.8M', 'Varies with device'). Requires cleaning before numeric use.
Installs	Character	Install count as a range (e.g., '10,000+', '1,000,000+'). Requires cleaning to use numerically.
Type	Character	Whether the app is Free or Paid.
Price	Character*	App price in USD (e.g., '\$4.99' or '0'). Stored as character — must be cleaned and converted.
Content Rating	Character	Age group the app is rated for (e.g., 'Everyone', 'Teen', 'Mature 17+').
Genres	Character	Semicolon-separated genre tags (e.g., 'Art & Design;Creativity'). Multi-value column.
Last Updated	Character	Date the app was last updated on the Play Store.
Current Ver	Character	Current version string of the app.
Android Ver	Character	Minimum Android version required (e.g., '4.0.3 and up').

** These columns look numeric but are stored as character strings due to formatting — they must be cleaned before analysis.*

Q1. Loading and Cleaning the Dataset

Load the dataset into R and examine its structure. Then answer the following:

```

1 # Load required libraries
2 library(tidyverse)
3
4 glimpse(GooglePlayStore)
5
6

```

21:1 (Top Level) ↕ R Script

```

R 4.5.2 ~ /
> # Load required libraries
> library(tidyverse)
> glimpse(GooglePlayStore)
Rows: 10,841
Columns: 13
$ App      <chr> "Photo Editor & Candy Camera & Grid & ScrapBook", "Coloring book moana", "U Launcher Lite ð€" FREE Live Cool Themes, Hide Apps",...
$ Category <chr> "ART_AND_DESIGN", "ART_AND_DESIGN", "ART_AND_DESIGN", "ART_AND_DESIGN", "ART_AND_DESIGN", "ART_AND_DESIGN", "ART_AND_DESIGN", "A...
$ Rating   <chr> "4.0999999999999996", "3.9", "4.7", "4.5", "4.3", "4.4000000000000004", "3.8", "4.0999999999999996", "4.4000000000000004", "4.7"...
$ Reviews  <dbl> 159, 967, 87510, 215644, 967, 167, 178, 36815, 13791, 121, 13880, 8788, 44829, 4326, 1518, 55, 3632, 27, 194216, 224399, 450, 65...
$ Size     <chr> "19M", "14M", "8.7M", "25M", "2.8M", "5.6M", "19M", "29M", "33M", "3.1M", "28M", "12M", "20M", "21M", "37M", "2.7M", "5.5M", "17...
$ Installs <chr> "10,000+", "500,000+", "5,000,000+", "50,000,000+", "100,000+", "50,000+", "50,000+", "1,000,000+", "1,000,000+", "10,000+", "1...
$ Type     <chr> "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", "Free", ...
$ Price    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ `Content Rating` <chr> "Everyone", "Everyone", "Everyone", "Teen", "Everyone", "Everyone", "Everyone", "Everyone", "Everyone", "Everyone", "Everyone", ...
$ Genres   <chr> "Art & Design", "Art & Design;Pretend Play", "Art & Design", "Art & Design", "Art & Design;Creativity", "Art & Design", "Art & D...
$ `Last Updated` <dtm> 2018-01-07, 2018-01-15, 2018-08-01, 2018-06-08, 2018-06-20, 2018-06-20, 2018-03-26, 2018-04-26, 2018-06-14, 2017-09-20, 2018-07-03, 2017-10...
$ `Current Ver`  <chr> "1.0.0", "2.0.0", "1.2.4", "Varies with device", "1.1000000000000001", "1", "1.1000000000000001", "6.1.61.1", "2.9.2", "2.8", "1...
$ `Android Ver`  <chr> "4.0.3 and up", "4.0.3 and up", "4.0.3 and up", "4.2 and up", "4.4 and up", "2.3 and up", "4.0.3 and up", "4.2 and up", "3.0 and...
>

```

a. (1 mark) Several columns that appear numeric are actually stored as characters. Convert the following columns to numeric format. Show the code you used.

- Reviews — remove any non-numeric characters and convert to numeric
- Price — remove the leading '\$' character and convert to numeric
- Ratings- remove any non-numeric characters and convert to numeric

```

8
9 google_selected <- GooglePlayStore %>%
10   select(Rating, Reviews, Type, Price, `Content Rating`, Category) %>%
11   mutate(
12     # Remove non-numeric characters from Reviews and convert to numeric
13     Reviews = as.numeric(gsub("[^0-9]", "", Reviews)),
14
15     # Remove "$" symbol from Price and convert to numeric
16     Price = as.numeric(gsub("\\$", "", Price)),
17
18     # Remove non-numeric characters from Rating and convert to numeric
19     Rating = as.numeric(gsub("[^0-9\\.]", "", Rating))
20   )
21
22 cat("\nAfter conversions:\n")
23 glimpse(google_selected)

```

20:4 (Top Level) ↕ R Script

```
Console Terminal x Background Jobs x
R 4.5.2 · ~/
+ Reviews = as.numeric(gsub("[^0-9]", "", Reviews)),
+
+ # Remove "$" symbol from Price and convert to numeric
+ Price = as.numeric(gsub("\\$", "", Price)),
+
+ # Remove non-numeric characters from Rating and convert to numeric
+ Rating = as.numeric(gsub("[^0-9\\.]", "", Rating))
+ )
>
> cat("\nAfter conversions:\n")

After conversions:
> glimpse(google_selected)
Rows: 10,841
Columns: 6
$ Rating      <dbl> 4.1, 3.9, 4.7, 4.5, 4.3, 4.4, 3.8, 4.1, 4.4, 4.1, ...
$ Reviews     <dbl> 159, 967, 87510, 215644, 967, 167, 178, 36815, ...
$ Type        <chr> "Free", "Free", "Free", "Free", "Free", "Free", ...
$ Price       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ `Content Rating` <chr> "Everyone", "Everyone", "Everyone", "Teen", "Ev...
$ Category    <chr> "ART_AND_DESIGN", "ART_AND_DESIGN", "ART_AND_DE...
> |
```

b. (1 marks) How many rows remain after removing records with missing, null, NaN or any other form of junk values in all of the following five columns of interest: Rating, Reviews, Type, Price, and Content Rating? What percent of rows were removed?

There are 9366 rows remaining after the cleaning process. One of these rows has a Content Rating labeled “Unrated.” If this value is considered invalid or junk data, the dataset would effectively contain 9365 valid rows. Overall, 13.61% of the original rows were removed during the cleaning step.

```

25 #Q.b
26 # Remove rows with missing/null/NaN values from all columns
27 google_clean <- google_selected %>% drop_na()
28
29 clean_rows <- nrow(google_clean)
30 total_rows <- nrow(GooglePlayStore)
31 rows_removed <- total_rows - clean_rows
32 percent_removed <- round((rows_removed / total_rows) * 100, 2)
33
34 cat("\nRows after cleaning:", clean_rows, "\n")
35 cat("Rows removed:", rows_removed, "\n")
36 cat("Percent removed:", percent_removed, "%\n")
37
38

```

```

> #Q.b
> # Remove rows with missing/null/NaN values from all columns
> google_clean <- google_selected %>% drop_na()
>
> clean_rows <- nrow(google_clean)
> total_rows <- nrow(GooglePlayStore)
> rows_removed <- total_rows - clean_rows
> percent_removed <- round((rows_removed / total_rows) * 100, 2)
>
> cat("\nRows after cleaning:", clean_rows, "\n")

Rows after cleaning: 9366
> cat("Rows removed:", rows_removed, "\n")
Rows removed: 1475
> cat("Percent removed:", percent_removed, "%\n")
Percent removed: 13.61 %
> |

```

Q2. Exploring App Categories

- a. **(1 mark)** How many unique app categories are in the dataset? Use the clean version of dataset from Q1-b:

There are 33 unique app categories in the dataset

```

37
38
39 #Q2.a
40 # Count the number of unique app categories
41 unique_categories <- google_clean %>%
42   distinct(Category) %>%
43   nrow()
44
45 cat("\nNumber of unique categories:", unique_categories, "\n")
46
47

```

```

>
> cat("\nNumber of unique categories:", unique_categories, "\n")

Number of unique categories: 33
>

```

b. (1 mark) Create a frequency table showing how many apps fall into each category. What are the top 3 most common categories?

The three most common app categories are FAMILY, GAME, and TOOLS.

```

47
48 #Q2.b
49 # Create a frequency table of app categories sorted by count
50 category_freq <- google_clean %>%
51   count(Category, sort = TRUE) %>%
52   mutate(Percentage = round((n / sum(n)) * 100, 2))
53
54 cat("\nFrequency Table - Categories:\n")
55 print(n = 33, category_freq)
56
57

```

Frequency Table - Categories:

```
> print(n = 33, category_freq)
```

```
# A tibble: 33 × 3
```

	Category	n	Percentage
	<chr>	<int>	<dbl>
1	FAMILY	1747	18.6
2	GAME	1097	11.7
3	TOOLS	734	7.84
4	PRODUCTIVITY	351	3.75
5	MEDICAL	350	3.74
6	COMMUNICATION	328	3.5
7	FINANCE	323	3.45
8	SPORTS	319	3.41
9	PHOTOGRAPHY	317	3.38
10	LIFESTYLE	314	3.35
11	PERSONALIZATION	314	3.35
12	BUSINESS	303	3.24
13	HEALTH_AND_FITNESS	297	3.17
14	SOCIAL	259	2.77
15	SHOPPING	238	2.54
16	NEWS_AND_MAGAZINES	233	2.49
17	TRAVEL_AND_LOCAL	226	2.41
18	DATING	195	2.08
19	BOOKS_AND_REFERENCE	178	1.9
20	VIDEO_PLAYERS	160	1.71
21	EDUCATION	155	1.65
22	ENTERTAINMENT	149	1.59
23	MAPS_AND_NAVIGATION	124	1.32
24	FOOD_AND_DRINK	109	1.16
25	HOUSE_AND_HOME	76	0.81
26	WEATHER	75	0.8
27	AUTO_AND_VEHICLES	73	0.78
28	LIBRARIES_AND_DEMO	65	0.69
29	ART_AND_DESIGN	62	0.66
30	COMICS	58	0.62
31	PARENTING	50	0.53
32	EVENTS	45	0.48
33	BEAUTY	42	0.45

```
> |
```

Q3. Summary Statistics for App Ratings

(2 marks) Calculate the following summary statistics for the Rating column (after removing NAs): mean, median, standard deviation, minimum, maximum, Q1, and Q3. Fill in the table below.

Statistic	Value
Min	1
Q1	4
Median	4.3
Mean	4.19
Q3	4.5
Max	5
Std Dev	0.515

```
57 #Q3
58 # Calculate summary statistics for the Rating column
59 rating_summary <- google_clean %>%
60   summarise(
61     Mean    = mean(Rating),
62     Median  = median(Rating),
63     SD      = sd(Rating),
64     Min     = min(Rating),
65     Max     = max(Rating),
66     Q1      = quantile(Rating, 0.25),
67     Q3      = quantile(Rating, 0.75),
68     IQR     = IQR(Rating)
69   )
70
71 cat("\nRating Summary Statistics:\n")
72 print(rating_summary)
73
74
```



```

> #Q3
> # Calculate summary statistics for the Rating column
> rating_summary <- google_clean %>%
+   summarise(
+     Mean    = mean(Rating),
+     Median  = median(Rating),
+     SD      = sd(Rating),
+     Min     = min(Rating),
+     Max     = max(Rating),
+     Q1      = quantile(Rating, 0.25),
+     Q3      = quantile(Rating, 0.75),
+     IQR     = IQR(Rating)
+   )
>
> cat("\nRating Summary Statistics:\n")

```

Rating Summary Statistics:

```

> print(rating_summary)
# A tibble: 1 × 8
  Mean Median    SD  Min  Max   Q1   Q3  IQR
  <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1  4.19    4.3 0.515    1    5    4    4.5  0.5
> |

```

Q4. Ratings by Content Rating Group

(2 marks) Calculate the mean and median rating broken down by Content Rating group (Everyone, Teen, Mature 17+, etc.). Fill in the table below.

Content Rating	Mean Rating	Median Rating	Count
Everyone	4.19	4.3	7420
Everyone 10+	4.26	4.3	397
Teen	4.23	4.3	1084

Mature 17+	4.12	4.2	461
Adults only 18+	4.3	4.5	3

```

74 #Q4.
75 ratings_by_content <- google_clean %>%
76   group_by(`Content Rating`) %>%
77   summarise(
78     Mean_Rating = mean(Rating),
79     Median_Rating = median(Rating),
80     Count = n()
81   ) %>%
82   arrange(desc(Count))
83
84 print(ratings_by_content)
85
86

```

```

> #Q4.
> ratings_by_content <- google_clean %>%
+   group_by(`Content Rating`) %>%
+   summarise(
+     Mean_Rating = mean(Rating),
+     Median_Rating = median(Rating),
+     Count = n()
+   ) %>%
+   arrange(desc(Count))
>
> print(ratings_by_content)
# A tibble: 6 × 4
  `Content Rating` Mean_Rating Median_Rating Count
  <chr>           <dbl>         <dbl> <int>
1 Everyone        4.19           4.3    7420
2 Teen            4.23           4.3    1084
3 Mature 17+      4.12           4.2     461
4 Everyone 10+    4.26           4.3     397
5 Adults only 18+ 4.3            4.5        3
6 Unrated         4.1           4.1        1
> |

```